

case is obviously not going to work out well. The simplest metric to check is if the height of the cooler is lesser than the chassis width. Some of the newer air-coolers have really tall designs which makes them incompatible with most of the budget cases. Aside from the height, checking for clearance regarding the other components, i.e. the optical disk drive and the hard drive which generally jut out from the drive cages should be a priority. As cases get smaller, the front-bay components such as the HDD and the ODD tend to come close to each other and can easily interfere with a large cooler. We'll explore case compatibility in greater detail in the next chapter.

TDP: Probably one of the most important aspects to check while buying a CPU cooler is whether it can actually cool your CPU. While manufacturers will rate their coolers generously, they don't always perform so good. The best way to figure out whether a cooler will fit the bill is to read reviews and see how well it compares with others in the market. Reviews allow you to sift through all the marketing mumbo-jumbo.

Size constraints: Heatsinks in air coolers are present in different sizes, from low-profile heatsinks to tall ones. Since the traditional heatsink isn't present in liquid coolers, they usually have a compact form factor. The size of your CPU cooler will hinder a few components including your RAM sticks, chassis, and the first PCI-e slot. Big heatsinks tend to take up space over the memory slots and if your RAM module has high-profile heatsinks, you won't be able to attach them. Beefier heatsinks might be tall enough to be wider than your chassis. Ensure that you check the dimensions of your cooler and chassis before making a decision otherwise you'll be forced to keep your side panel open. Finally, just like the way the heatsink might extend to the RAM slots horizontally, it could also do the same to cover up the first PCI-e slot. If you're installing closed-loop or open-loop liquid coolers, you have to consider the length of the tubes to make sure it reaches



Credit: Anandtech

Many coolers of different sizes.